

1. Definition : IAM (Identity and Access Management) is an AWS service used to securely control access to resources by managing users, permissions, and roles.

2. IAM Overview

- **Provides authentication (who can login).**
- **Provides authorization (what actions they can perform).**
- **Allows fine-grained access control.**
- **Works globally (not region specific).**

3. IAM Policies

- **Policies are JSON documents that define permissions.**
- **They specify Allow or Deny access to AWS resources.**

Structure of Policy:

- **Effect (Allow/Deny)**
- **Action (e.g., S3:Read)**
- **Resource (specific AWS resource)**

4. Types of Policies

- **Managed Policies (predefined by AWS)**
- **Custom Policies (user-created)**
- **Inline Policies (attached to single user/group)**

5. Example : A policy that allows a user to read files from S3 bucket only.

IAM Users

1. Definition : IAM User is an individual identity created in AWS for a person or application. Each user has unique login credentials and permissions. It allows secure access to AWS services based on assigned policies. IAM users help in managing separate access for different individuals.

2. Features

- **Each user has unique credentials.**
- **Can have password (console access).**
- **Can have access keys (programmatic access).**

3. Permissions

- **Permissions are assigned using policies.**
- **Can be assigned directly or via groups.**

4. Types of Access

- **AWS Management Console**
- **CLI (Command Line Interface)**
- **API access**

5. Example : A company creates different IAM users for developer, tester, and admin with different permissions.

6. Advantages

- **Better security control**
- **Easy user management and Separate access for each individual**

1. Definition

IAM Group is a collection of IAM users that share the same permissions. It simplifies access management by assigning permissions to the group instead of individual users. Users added to a group automatically inherit its permissions. It helps in organizing users based on roles like developers or admins.

2. Features

- **Policies are attached to groups instead of individual users.**
- **Users in the group inherit permissions automatically.**

3. Benefits

- **Easy management of multiple users**
- **Reduces duplication of policies**
- **Improves security control**

4. Example : A “Developers” group has access to EC2 and S3. All developers added to the group get same access.

Access Key & Secret Access Key

1. Definition : Access Key and Secret Access Key are security credentials used for programmatic access to AWS services. They allow applications and users to interact with AWS using API, CLI, or SDKs. The access key acts as a public identifier, while the secret key acts as a private password. These keys must be handled securely to prevent unauthorized access.

2. Components

- **Access Key ID (public)**
- **Secret Access Key (private)**

3. Usage

- **Used in AWS CLI, SDKs, APIs**
- **Helps applications interact with AWS services**

4. Working

- **Access key identifies user**
- **Secret key authenticates the request**

5. Security Practices

- **Never share secret key**
- **Rotate keys regularly**
- **Store securely (e.g., environment variables)**

6. Advantages

- **Enables automation**
- **Supports integration with applications**

7. Example

A mobile app uses access keys to upload files to S3 bucket programmatically.

1. Definition

MFA (Multi-Factor Authentication) is a security mechanism that requires more than one method of authentication. It combines something the user knows (password) with something the user has (OTP/device). This adds an extra layer of protection to AWS accounts. It helps prevent unauthorized access even if the password is compromised.

2. Types of Authentication

- **Password (something you know)**
- **OTP from device (something you have)**

3. Working

- **User enters password**
- **Then enters OTP generated from device/app**

4. Devices Used

- **Mobile authenticator apps (Google Authenticator)**
- **Hardware MFA devices**

5. Advantages

- **Extra layer of security**
- **Protects against unauthorized access**

6. Example

Even if password is stolen, attacker cannot login without OTP.

IAM Report

1. Definition

IAM Report (Credential Report) is a document that provides details about all IAM users and their credentials. It includes information such as password status, access keys, and MFA usage. This report helps administrators to monitor security and user activity. It is useful for auditing and maintaining a secure AWS environment.

2. Purpose

- **Helps monitor security status**
- **Identifies unused or weak credentials**

3. Information Included

- **Password usage**
- **Access key status**
- **MFA enabled or not**
- **Last login time**

4. Uses

- **Security auditing**
- **Compliance checking**
- **Identifying inactive users**

5. Example

Admin checks report to find users without MFA enabled and improves security.

1. Definition

Amazon EC2 (Elastic Compute Cloud) is a web service provided by AWS that allows users to create and run virtual servers in the cloud.

It provides scalable computing capacity so users can quickly launch and manage servers.

EC2 eliminates the need to invest in physical hardware.

It is widely used for hosting applications, websites, and services.

2. Key Features

- **On-demand virtual machines (instances)**
- **Scalable resources (CPU, RAM, storage)**
- **Pay-as-you-go pricing**
- **Multiple operating systems support (Linux, Windows)**

3. Components

- **Instance (virtual server)**
- **AMI (Amazon Machine Image)**
- **Instance Type (CPU, memory configuration)**

4. Benefits

- **High flexibility**
- **Cost-effective**
- **Easy to deploy and manage**

5. Example

A company hosts its website on EC2 instead of buying physical servers.

Elastic Block Storage (EBS)

1. Definition : Elastic Block Storage (EBS) is a storage service provided by AWS that offers persistent block-level storage for EC2 instances. It works like a virtual hard disk attached to an EC2 instance. Data stored in EBS remains safe even after the instance is stopped. It is mainly used for storing operating systems, databases, and application data.

2. Features

- **Persistent storage**
- **High availability and durability**
- **Can be attached/detached from EC2**

3. Types of EBS Volumes

- **SSD (for high performance)**
- **HDD (for large storage)**

4. Advantages

- **Reliable data storage**
- **Backup using snapshots**
- **Suitable for databases**

5. Example

An EC2 instance uses EBS to store a MySQL database, and data remains safe even after restart.

1. Definition

Amazon Machine Image (AMI) is a pre-configured template used to launch EC2 instances in AWS.

It includes operating system, application software, libraries, and configurations required for a server. AMI enables users to quickly create multiple identical instances. It ensures fast deployment and consistency in cloud environments.

2. Components of AMI

- **OS (Linux/Windows)**
- **Application software**
- **Launch permissions**
- **Block device mapping (storage details)**

3. Types of AMI

- **Public AMI is available for all users.**
- **Private AMI is used within an organization.**
- **Marketplace AMI provides paid services.**

4. Features

- **It allows quick instance launching.**
- **It provides consistent environment.**
- **It supports customization.**

5. Advantages

- **It reduces setup time.**
- **It ensures uniform configuration.**
- **It supports scalability.**

6. Example

A company creates an AMI with pre-installed web server and launches multiple EC2 instances.

7. Conclusion

AMI simplifies deployment and scaling of cloud servers.

1. Definition

Instance purchasing options define different pricing models for using EC2 instances. They allow users to choose resources based on cost, duration, and workload type. AWS provides flexible options for both short-term and long-term usage. It helps in optimizing cloud costs.

2. Types of Purchasing Options

- **On-demand instances are used for flexible short-term usage.**
- **Reserved instances provide lower cost for long-term commitment.**
- **Spot instances offer very low cost but can be interrupted.**
- **Dedicated hosts provide physical servers for single customer.**

3. Features

- **It provides flexible pricing models.**
- **It supports different workload requirements.**
- **It helps manage cloud budget.**

4. Advantages

- **It reduces overall infrastructure cost.**
- **It provides better cost control.**
- **It allows efficient resource utilization.**

5. Example

A startup uses spot instances for batch processing to reduce cost.

6. Conclusion

Purchasing options help in cost-effective cloud usage.

1. Definition

EC2 Instance Types define the hardware configuration of virtual machines such as CPU, RAM, and storage.

Security Group is a virtual firewall that controls network traffic.

Both are essential for performance and security of EC2.

They ensure proper resource allocation and secure access.

2. EC2 Instance Types

- **General purpose instances** provide balanced performance.
- **Compute optimized instances** are used for CPU-intensive tasks.
- **Memory optimized instances** are used for RAM-intensive applications.
- **Storage optimized instances** are used for large storage needs.

3. Security Group Features

- **It controls inbound and outbound traffic.**
- **It works as a virtual firewall.**
- **It uses rules based on port, protocol, and IP address.**
- **It is stateful (response traffic is automatically allowed).**

4. Advantages

- **It improves performance selection.**
- **It ensures secure communication.**
- **It provides flexible access control.**

5. Example

Security group allows HTTP (port 80) and SSH (port 22) access.

6. Conclusion

Instance types and security groups ensure efficient and secure cloud operation.

1. Definition

IP address is used to identify EC2 instances in a network.

Public IP allows communication over the internet, while Private IP is used within AWS internal network (VPC).

Both types of IPs are important for communication and security.

They help separate internal and external access.

2. Public IP

- **Public IP is used to access instance from internet.**
- **It is dynamically assigned.**
- **It changes when instance is restarted.**

3. Private IP

- **Private IP is used within VPC.**
- **It remains constant for instance.**
- **It is more secure.**

4. Features

- **It supports internal and external communication.**
- **It improves network management.**

5. Advantages

- **It provides secure internal communication.**
- **It controls public access.**

6. Example

Web server uses public IP and database server uses private IP.

7. Conclusion

IP addressing ensures secure and efficient communication.

1. Definition

Amazon EBS is a persistent block storage service for EC2 instances.

Snapshot is a backup of EBS volume stored in S3.

Snapshots are incremental and store only changed data.

They help in data backup and recovery.

2. Features

- **EBS provides persistent storage.**
- **Snapshot provides backup facility.**
- **Snapshot stores only changed data.**
- **It supports easy restore.**

3. Advantages

- **It protects data from loss.**
- **It supports disaster recovery.**
- **It ensures high durability.**

4. Example

Snapshot is taken before software upgrade.

5. Conclusion

EBS and snapshots provide reliable storage and backup solution.

AWS CLI

1. Definition

AWS CLI is a tool used to manage AWS services using command line interface.

It allows users to interact with AWS without using GUI.

It supports automation using scripts.

It is useful for efficient cloud management.

2. Features

- **It allows command-based access.**
- **It supports scripting and automation.**
- **It works across multiple platforms.**

3. Advantages

- **It saves time.**
- **It is faster than GUI.**
- **It is useful for repetitive tasks.**

4. Example

User launches EC2 instance using CLI command.

5. Conclusion

AWS CLI provides fast and automated cloud operations.

1. Definition

Bootstrap script is a script that runs automatically when an EC2 instance starts.
It is used to install software and configure the system.
It helps automate setup process.
It reduces manual configuration effort.

2. Features

- **It runs at instance launch.**
- **It installs required software automatically.**
- **It supports automation.**

3. Advantages

- **It saves time.**
- **It ensures consistent setup.**
- **It reduces human errors.**

4. Example

Install Apache server automatically at startup.

5. Conclusion

Bootstrap script simplifies instance configuration process.

Elastic Load Balancing

1. Definition

Elastic Load Balancing (ELB) is a service that distributes incoming traffic across multiple EC2 instances.

It ensures high availability and fault tolerance.

It automatically handles traffic load.

It improves performance and reliability of applications.

2. Types of ELB

- **Application Load Balancer handles HTTP/HTTPS traffic.**
- **Network Load Balancer handles high-performance traffic.**
- **Classic Load Balancer provides basic load balancing.**

3. Features

- **It distributes traffic automatically.**
- **It detects unhealthy instances.**
- **It supports scaling.**

4. Advantages

- **It prevents server overload.**
- **It improves availability.**
- **It ensures fault tolerance.**

5. Example

Traffic is divided between multiple servers to avoid failure.

6. Conclusion

ELB ensures efficient traffic management and high availability.

1. Definition

Auto Scaling is a service that automatically adjusts the number of EC2 instances based on demand.

It increases instances during high traffic and decreases during low traffic.

It ensures application performance and availability.

It also helps in cost optimization.

2. Features

- **It automatically increases or decreases instances.**
- **It works based on defined policies.**
- **It maintains system performance.**

3. Advantages

- **It ensures high availability.**
- **It reduces cost.**
- **It improves performance.**

4. Example

More instances are added during heavy website traffic.

5. Conclusion

Auto Scaling provides efficient resource management and cost control.